Chair travel case

ceremonial chairs · build plans · Rev G (CNC-cut kit)

Case outside	1018 W × 1008 D × 1761 H mm on castors
Inside	1000 × 990 × 1605 mm, lined 20 mm EPE, 20 mm EVA floor
Chair	927 × 920 × 1547 (maker's STEP, 9 Nov), 15–18 mm air gap all round
Build	9 mm black hexa ply walls, top and lid on a 17 mm formply base; CNC-cut panels, glued + pinned, 30 mm single angle, 210 mm deep lift-off lid on a hybrid joiner split
Access	5 butterfly latches; tip the lid's top out, lift it 70 mm clear of the base corners; chair slides out forward
Mobility	100 mm castors (100 kg each, 2 braked) bolted to the base at the corners, 6 handles
Weight	about 81 kg empty, 134 kg packed (calculated: weigh the first one)

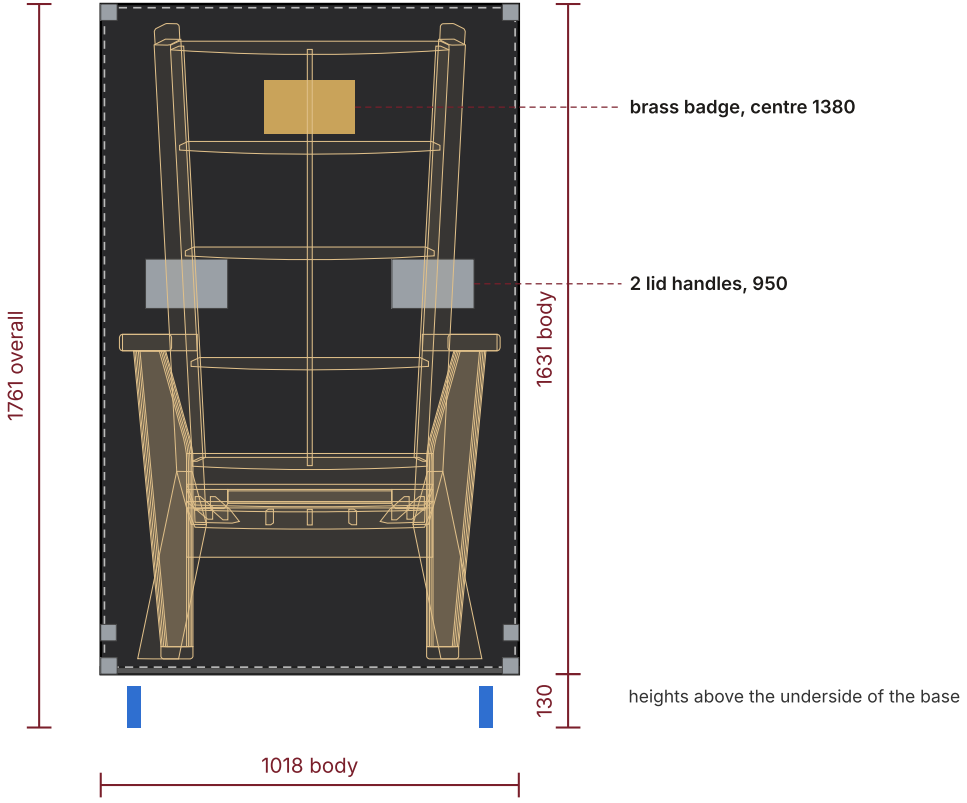
Before the final CNC files go: confirm the real ply thickness, and mate the hybrid joiner on two offcuts to check its 4 mm gap. Set foot channels on the first finished chair. Chair geometry is maker's STEP model. Badge needs the client' official vector logo.



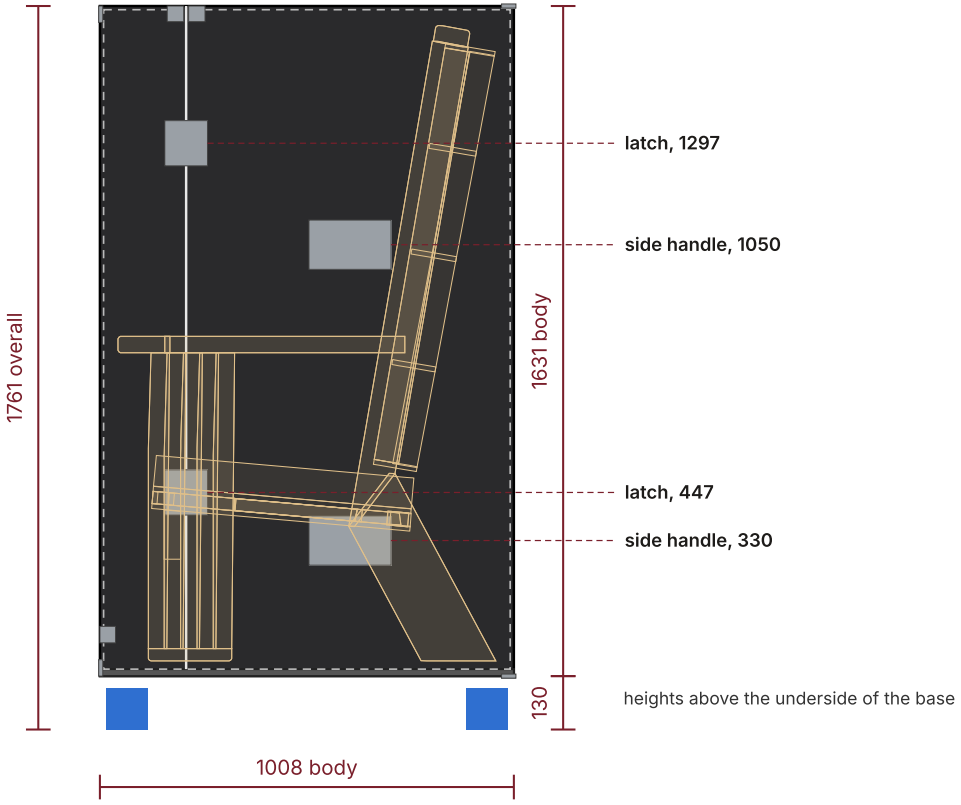


Elevations

FRONT (lid on), chair outlined

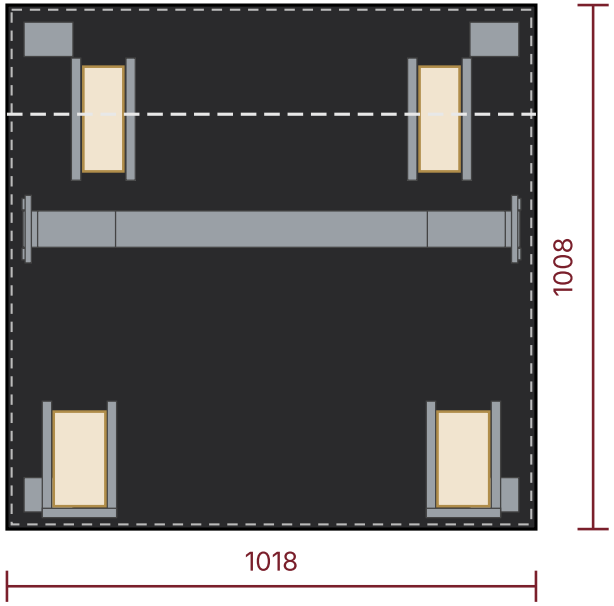


LEFT SIDE, lid at left (split 210 back)

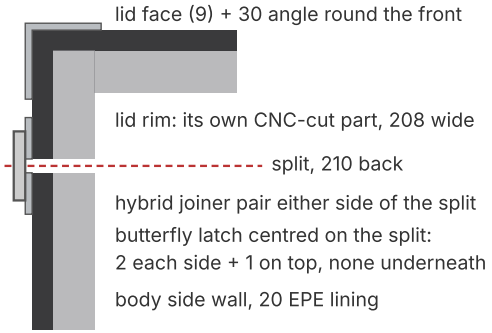


Plan and details

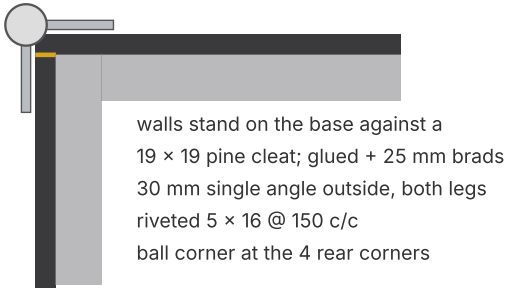
PLAN at floor, lid at top



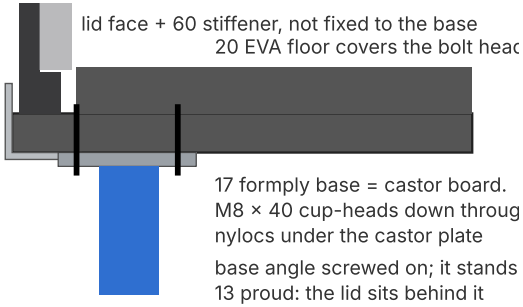
A Lid and split (plan section through a side wall)



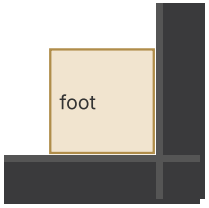
B Rear corner (plan section)



C Base, lid lip and castor (section at the front)

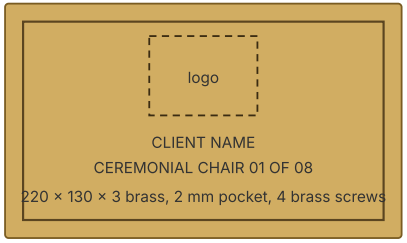


D Rear foot channel (plan)

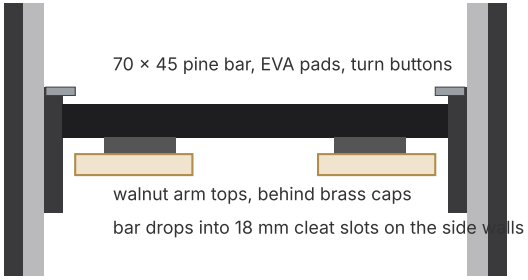


Channel either side of every foot. Front channels open front and back; rear feet get a back stop. Set on the first chair.

E Brass badge

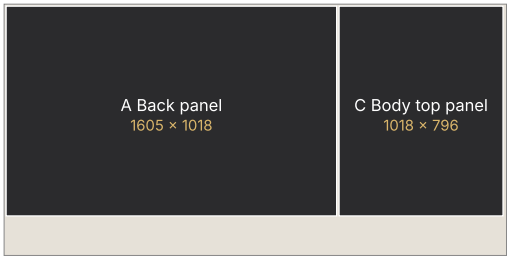


F Hold-down bar (section across case)

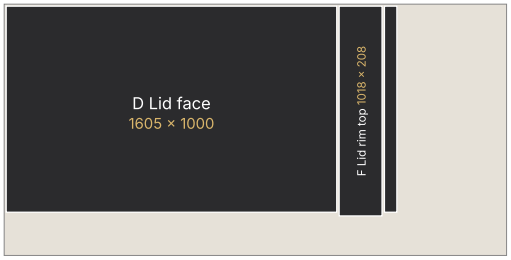


Cut list · hexa ply, 4 sheets per case

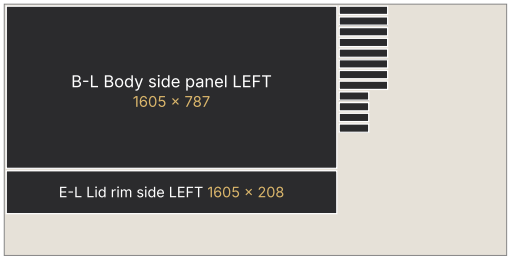
Sheet 1



Sheet 2



Sheet 3



Sheet 4



CNC parts, one case

Part	Name	Size	Qty
A	Back panel	1018 × 1605	1
B-L	Body side panel LEFT (mirror of B-R)	787 × 1605	1
B-R	Body side panel RIGHT	787 × 1605	1
C	Body top panel	1018 × 796	1
D	Lid face	1000 × 1605	1
E-L	Lid rim side LEFT (mirror of E-R)	208 × 1605	1
E-R	Lid rim side RIGHT	208 × 1605	1
F	Lid rim top	1018 × 208	1
G	Lid bottom stiffener	1000 × 60	1
H	Foot channel strip, front	235 × 40	8
J	Foot channel strip, rear	204 × 40	8
K	Rear foot back-stop strip	143 × 40	4
L	Bar slot cleat strip	180 × 22	8
M	Base	1018 × 1008	1

Panels are CNC cut by the supplier from the Rev G DXF pack, with handle openings, latch notches, the badge pocket and castor marks. Left and right are mirror images. Confirm the real sheet thickness and the 4 mm hybrid joiner gap before the final files go. Extrusion runs: 30 mm single angle 958, 958, 948, 1571, 743, 1495, 155, 948, 1571, 743, 1495, 155, 958, 958; hybrid joiner 1575, 1575, 968, 1575, 1575, 968.

Build order

1. **Order and collect.** Panels: order the CNC-cut kits from the chosen panel supplier (quote requests are drafted for Titan AV, Armor Cases, DMK and Just Routing). Before the final files go, confirm the real sheet thickness and buy one pair of hybrid joiner lengths and one latch from Titan so the split gap and latch notches can be checked. Hardware, extrusion and foam: Titan AV, 244 Earnshaw Rd, Northgate (order online for pickup, ask for uncut 3 m extrusion lengths). Pine, bolts, screws, glue, contact spray and paint from Bunnings. Send the logo artwork to whoever makes the chair brass for the badges.
2. **Check the kit.** Lay out one case's parts against the parts list (A to M). Check the hexa face is on the outside of both side panels and both lid rims: left and right are mirror images. Offer a handle and a latch up to their openings. Mate the two hybrid joiner halves on two offcuts and check the gap they hold between the ply edges matches the 4 mm the panels were cut for.
3. **Prep the parts.** Paint the cut edges of the base black. Glue the cleat strips in pairs (H, J, K, L) to make 18 mm cleats and stick felt on the faces that will touch the chair. Cut the three 19 × 19 pine cleats and the 70 × 45 hold-down bar.
4. **Glue up the body and the lid.** Body: glue and screw the three pine cleats to the base, 9 mm in from the back and side edges. Stand the back (A) and the two sides (B-L, B-R, notched edges to the front) on the base against the cleats; glue (PU) and pin with 25 mm brads, then screw through each wall into its cleat every 200 mm, 9 mm above the base. Glue and pin the top (C) on, notch to the front. Lid: glue and pin the two rims (E-L, E-R) to the edges of the face (D), the top strip (F) over all three, and the 60 mm stiffener (G) inside the bottom of the face. Nothing on the lid is glued to the base. Check both for square before the glue goes off.
5. **Fit the split frame.** Fit the hybrid joiner pair up both sides and across the top of the split, lid half on the lid and body half on the body, mitred at the two top corners, starting 14 mm above the base so they clear the lip. The joiner bridges the latch notches: cut it away at each notch to match (jigsaw or router with the notch as the guide). Rivet at 150 mm. Stand the lid on the base and check it seats all round with the latch notches lined up.
6. **Edges and corners.** 30 mm single angle on the base perimeter (screwed with 8g x 16 wafer-head screws: the base is too thick to rivet), the two rear verticals, the top back and sides, and round the front of the lid. Rivet into the 9 mm ply at 150 mm with 5 × 16 rivets and washers, drilled about 5.1 mm. The base angle stands 13 mm above the base along the front and the first 210 mm of each side: that lip is what the lid sits behind. Ball corners on the four rear corners; flat corners on the lid's two top front corners and on the base's two front corners (screwed to the base only: their 80 mm legs stand in front of the lid's bottom corners and keep it in). The lid's two front verticals start above those corners: one full 1.5 m bar each, measured down from the top mitre, with a 90 degree brace plate on the lid over the short gap below. Brace plates also go either side of the split on each top edge. With 1.5 m bars, the two rear verticals and the four hybrid verticals each need a 70 to 75 mm filler at the bottom; with uncut 3 m lengths they don't.
7. **Latches and handles.** Five butterfly latches straight into the pre-cut notches on the split: two each side and one on top, dish half on the body, catch half on the lid. Six recessed handles into the pre-cut openings: two each side, two on the lid face. Rivet through the hardware's own holes.
8. **Castors.** The base has the four castor plate centres marked, 80 mm in from the sides and 66 mm in from the front and back. Centre each plate on its mark, drill through its holes, and bolt with four M8 × 40 cup-head bolts, heads inside (they end up under the floor mat), nyloc nuts under the plate. Braked pair at the lid end.
9. **Hold-down bar slots.** Before lining, glue and screw two 18 mm cleats (L) to each side wall to form a vertical slot centred about 220 mm back from the split edge, with the bottom at arm-top height. Fit a turn button at the top of each slot. Paint the 70 × 45 bar black and glue the two EVA pads under it.
10. **Line it.** Cut the foam to the built box, not the drawing. Spray-contact 20 mm EPE on the back, sides and top of the body (starting above the cleats) and on the lid face and lid rims; cut round the latch and handle pans. Then the 20 mm EVA floor mat between the linings: the feet stand on it.
11. **First-chair fit.** Stand the first finished chair in, centred, 15 mm off the lid foam. Mark round the four feet. Glue and screw the 18 mm cleats (H front, J rear, felt faces) 4 mm off each foot, through the mat into the base: a channel either side of every foot, open front and back on the front feet so the chair slides out, plus a back stop (K) behind each rear foot. Drop in the padded hold-down bar so its pads sit on the walnut arm tops behind the brass caps, and turn the buttons over it. Nothing touches leather or brass.
12. **Badge.** Fit the engraved brass plate into its pre-cut 2 mm pocket on the lid face with four brass countersunk screws.
13. **Test and record.** Roll it, brake it, push it from the 1050 handle (not the top) and tilt it 10 degrees. Take the lid off and on a few times: unlatch, tip the top out a finger-width to free the top joint, lift it 70 mm clear of the base corners, walk it away. Load and unload the chair twice. Weigh it packed and note the weight for freight. Photograph it for the handover.

Handling: about 134 kg packed. Tailgate truck or forklift, two people. Brake the castors and strap to the pallet for freight. Not stackable: mark the top. Needs about 1.05 m clear width.